

forms a simple voltage follower (its output follows its input), providing a low output impedance to drive into the standard power amplifier.

Assemble the circuit on a general-purpose PCB and fit it to the rear of an

amplifier. The unit must be compact, yet robust. So use a very sturdy aluminium extrusion for the cabinet in order to neatly house the assembled PCB.

To ensure simple operation, there are only three connections to the unit.

First, mains power is tapped from the transformer. The second lead carries the 9V output to the amplifier. The third is the guitar signal input at the five-way socket for connection to the effect pedal. ●

TACHOMETER

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A tachometer is an instrument that measures the rotational speed of a shaft or disk in a motor or other machine. Here we present the basic version of the tachometer

that shows the revolutions per second (RPS) on a digital display.

Fig. 1 shows the block diagram of the tachometer. Fig. 2 shows the circuit of the tachometer, which

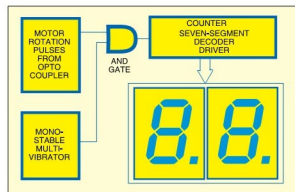


Fig. 1: Block diagram of the tachometer

is built around timer IC 555 (IC1), IC 7811 (IC2), IC CD4081 (IC3), IC CD4069 (IC4), two CD4033 ICs (IC5 and IC6) and two common-cathode displays (each LT5543) along with some discrete components. IC1 is wired in monostable mode and produces 1-second pulse when triggered by switch S2. Variable

resistor VR1 is used for setting the time period of IC1. IC2 generates pulses when the shaft is moving.

IC5 (CD4033) is a decade counter with seven-segment decoded output, which is connected to DIS1 for displaying unit's place. Pins 2 and 14 of IC5 are grounded. Pin 5 (CO) of IC5 provides a pulse when the count overflows (count reaches from 0 to 9). This pulse is used as the clock input to IC6 via NOT gate N3. That is, after ten pulses from IC5, there will be an increment of one pulse in IC6. Thus when the unit's counting completes in DIS1, further counting is shown on DIS2. A maximum of 99 digits can be counted and shown on DIS1 and DIS2.

Working of the circuit is simple. Place IC2 such that when the motor shaft rotates, it cuts the IR signal to produce pulses at its output pin 3. That is, as the motor rotates, the shaft

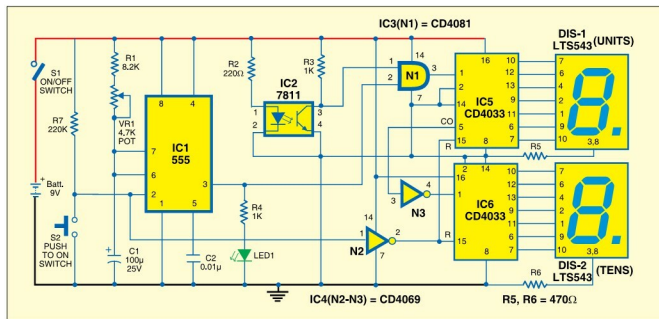


Fig. 2: Circuit of the tachometer